

Lotka—Volterra modeling of a spatially informed three-species microbial community as a two-species co-culture perturbed by the third

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Table of contents

1	Abstract	2
2	Introduction	2
2.1	Importance of community and habitat complexity in microbial ecology	2
2.2	The simple yet powerful framework of the Lotka—Volterra equation	3
3	Methods and Data	3
3.1	Experimental methods	3
3.2	Data processing and Lotka—Volterra parameter estimation	4
3.3	Software	5
4	Results	5
4.1	<i>P. putida</i> benefits from and outcompetes <i>B. subtilis</i> and is itself outcompeted by <i>T. viride</i>	5
4.2	Parametrization of Lotka—Volterra interaction coefficients suggests shift in intra- and inter-specific competition with increased spatial complexity	7
5	Discussion	8
5.1	Poor fit of the <i>B. subtilis</i> monoculture	9
5.2	Interpretation of interaction coefficients	9
6	Statement of authorship, contributions and use of LLMs	10
7	References	10
8	Supplementary Material	11

1 Abstract

This project implements a Lotka—Volterra model to analyse time series growth data, derive interaction coefficients and explore how a three-species interaction can be modeled from limited growth data of only two of the three species framing the third species as a perturbation to the interaction of the others. Fluorescently labeled *Pseudomonas putida* and *Bacillus subtilis* bacteria were co-cultured with a non-labeled *Trichoderma viride* fungus in microfluidic mazes with different degrees of fractal complexity for 22 days. Logistic growth curves, growth rates (r), carrying capacities (K) and interaction coefficients ($a_{ii}, a_{jj}, a_{ij}, a_{ji}$) were estimated with the R package **gauseR**. *B. subtilis* promoted and *T. viride* inhibited the growth of *P. putida*, while *B. subtilis* was strongly suppressed by both. The LV model fitted all conditions well except the *B. subtilis* monoculture. The promoting effect of *B. subtilis* on *P. putida* was reflected in the coefficients, and in the three-species culture this effect shifted from inhibitory to promoting at the highest spatial complexity. However, all coefficients were close to zero, the suppression of *B. subtilis* by *P. putida* and *T. viride* was not captured, and the effect of the unmeasured fungus is absorbed into effective coefficients of the two measured species. A simple Lotka—Volterra model thus reproduces the main growth patterns of *P. putida* but offers limited mechanistic insight into the three-species interactions.

2 Introduction

2.1 Importance of community and habitat complexity in microbial ecology

In microbial ecology, most experimental research is focused on either community response to a generalized change in ecosystem properties (e.g. temperature, pH, nutrient availability, toxins) by characterizing communities on a very broad and data-generation driven “-omics” approach with limited reconnaissance for functional and mechanistic processes (Faust and Raes 2012), or it focuses on two species interactions — of oftentimes synthetic consortia — with limited transferability to *in situ* complexity, neglecting the emergence of ecosystem properties and higher order interactions in microbial interaction networks (Levine et al. 2017).

Moreover, most of this culturing-technique based approaches is based entirely on well-mixed conditions. This poses great challenges for the study of community driven effects with relevance of the spatial arrangement like biofilm formation, quorum sensing or niche differentiation (Orr et al. 2025; Nadell et al. 2016). This is especially limiting for the study of soil habitats, since their pore space is characterized by a very high spatial complexity and thus microhabitat diversity.

P. putida and *B. subtilis* are frequently co-isolated from soil samples and *T. viride* is also a common member of soil microbial communities. Therefore all three species are thought to be stably coexisting in soil habitats under natural conditions. The central question of these experiments are how *T. viride* affects the growth of *P. putida* and *B. subtilis* facilitating or

hindering coexistence, by suppressing or supporting one or both of the species or by altering the interactions between them.

2.2 The simple yet powerful framework of the Lotka—Volterra equation

The Lotka-Volterra (LV) equations (Lotka 1925; Volterra 1926) are while being a mathematically relatively simple differential equation still a very powerful approach in ecology that can represent very complex, multi-layered interaction networks without requiring a complete mechanistic understanding of these interactions on a molecular level (Mühlbauer et al. 2020). The basic LV equation employed throughout this study describes the changes per unit time in the abundance of species N_i as

$$\frac{dN_i}{dt} = N_i(r_i + a_{ii}N_i + \sum_j (a_{ij}N_j)) \quad (1)$$

with r_i as the intrinsic growth rate, a_{ii} being the intra-specific competition (i.e. the effect of species i on itself, a_{ij} describing the effect of species j on species i and N_j as the abundance of species j .

Here, the interaction dynamics of the three-species system are approximated with this simple model to test whether the experimental data can already be fitted satisfactorily at this level of complexity, or whether more sophisticated models are needed. Fitting an LV model can also indicate whether the interactions are consistent with simple pairwise competition or whether higher-order interactions are likely at work (Grilli et al. 2017). In this experimental setup, time series growth data are available for only two of the three species. The overall aim is therefore to assess whether an LV model can represent the data adequately, and whether this approach is a viable way to gain mechanistic insight into a three-species system in a spatially structured habitat.

3 Methods and Data

3.1 Experimental methods

The wet lab experiments consist of semi-continuous culturing on microfluidic chips containing micro-scale mazes of different fractal complexity (f0-f5, Hilbert curve space filling design, Hilbert (1935)) and normalized internal volume. Each chip comprises a culture/media inoculation area from where mazes branch off Figure 1.

P. putida and *B. subtilis* carry a genomically integrated expression cassette for green (GFP) and red (RFP) fluorescent protein, respectively. Fluorescence intensity levels are used as a proxy for species abundance throughout this study. Culturing experiments were maintained for

over 22 days with a 48 h media-replenishment cycle approximating close-to unlimited nutrient supply (M9 minimal medium). In the same cycle, the mazes were imaged by epi-fluorescence microscopy collecting spatially resolved GFP and RFP signals. Sample size per time point was $n = 28$ for most conditions, $n = 7$ for the *B. subtilis*-only condition, and $n = 35$ for some time points of the *T. viride* + *P. putida* (TP) and three-species (TPB) conditions.

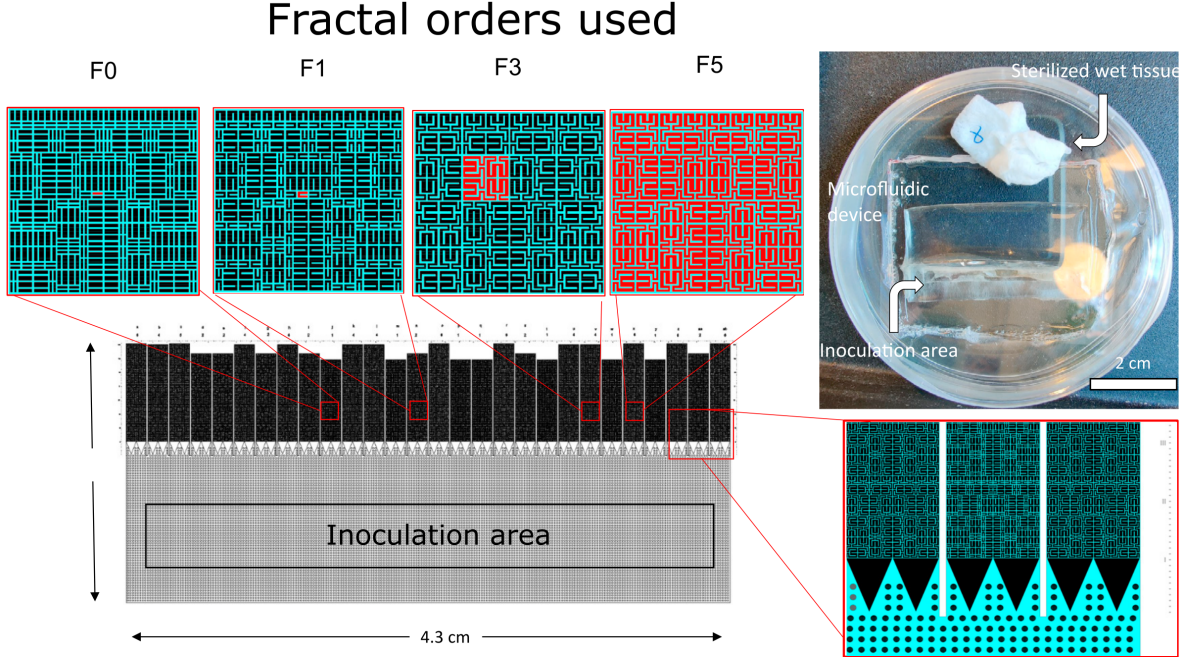


Figure 1: Fractal chip simulating microhabitats of different physical complexity. Figure kindly provided by Arellano-Caicedo et al. (2023).

3.2 Data processing and Lotka—Volterra parameter estimation

Raw microscopy images were aligned and background-subtracted in ImageJ using a rolling-ball algorithm with the ball diameter set to the channel width of the mazes. Fluorescence intensity was selectively measured from the maze areas only. A combined data frame across all time points and conditions was built in R, and means, standard deviations and standard errors were calculated per condition, fractal level, time point and fluorescence channel. Channels were assigned to species (GFP to *P. putida*, RFP to *B. subtilis*), and only the channel(s) of species actually present in a condition were retained. Sampling time points were converted to days after inoculation, and the data were arranged into a long format that allowed every condition $\times$ fractal level combination to be fitted automatically.

Each combination was fitted with the `gause_wrapper` function of the `gauseR` library, which works in two steps. First, time-lagged abundances are generated with `get_lag` and used to

calculate the per-capita growth rate. For a single species, the logistic model defines it as

$$\frac{dN}{dt}/N = r - a_{ii}N \quad (2)$$

with r the intrinsic growth rate and a_{ii} the self-limitation. Per-capita growth rate is regressed linearly against abundance (for several species, against the abundances of all species), so that r is the y-intercept, a_{ii} the slope, and the carrying capacity K the x-intercept (i.e. $K = -r/a_{ii}$). These linearized estimates serve as starting values. Second, the starting values (and initial abundances) are refined by a numerical optimizer (`optim`) that works in log-space, with parameter signs fixed to those of the linear estimates. The optimizer repeatedly solves the LV system with an ODE solver and minimizes the squared error between simulated and observed, standardized abundances. Rough parameter uncertainties (± 1 SD) are derived from the Hessian of the optimization.

Model fit was quantified per condition, fractal level and species. The fitted curve was interpolated at the observed time points, and goodness of fit was expressed as $R^2 = 1 - SS_{\text{res}}/SS_{\text{tot}}$ (which can become negative for fits worse than the mean) and as the root mean squared error (RMSE) in fluorescence units. Parameters were re-indexed to a fixed species order so that, for example, a *B. subtilis* monoculture reports a_{22} rather than a_{11} .

3.3 Software

Analyses were run in R 4.6.1 with `gauseR` 1.3. Data wrangling, plotting and table generation used `tidyverse`, `ggh4x`, `kableExtra` and `readxl`. Image processing was done in ImageJ 1.54, and the report was written in RStudio 2026.09.0 (Build 174) and rendered with Quarto 1.10.18, pandoc 3.7.0.2 and LuaHBTeX 1.24.0 (TeX Live 2026).

4 Results

4.1 *P. putida* benefits from and outcompetes *B. subtilis* and is itself outcompeted by *T. viride*

Figure 2 shows the time series of fluorescence intensity, which serves as a proxy for species abundance, for all mono- and co-culture combinations in the f3 maze. No growth data exist for *T. viride*. The presence of *B. subtilis* promotes the growth of *P. putida*, whereas *T. viride* inhibits it. *B. subtilis*, in turn, is strongly inhibited by both *P. putida* and *T. viride*, but the inhibition by *T. viride* is stronger and leads to essentially no detectable growth of *B. subtilis*. Even so, co-inoculation with *B. subtilis* still enhances *P. putida* growth. In the three-species culture, the promoting effect of *B. subtilis* cannot compensate for the inhibition by *T. viride*, so the growth pattern resembles that of the *P. putida* + *T. viride* co-culture.

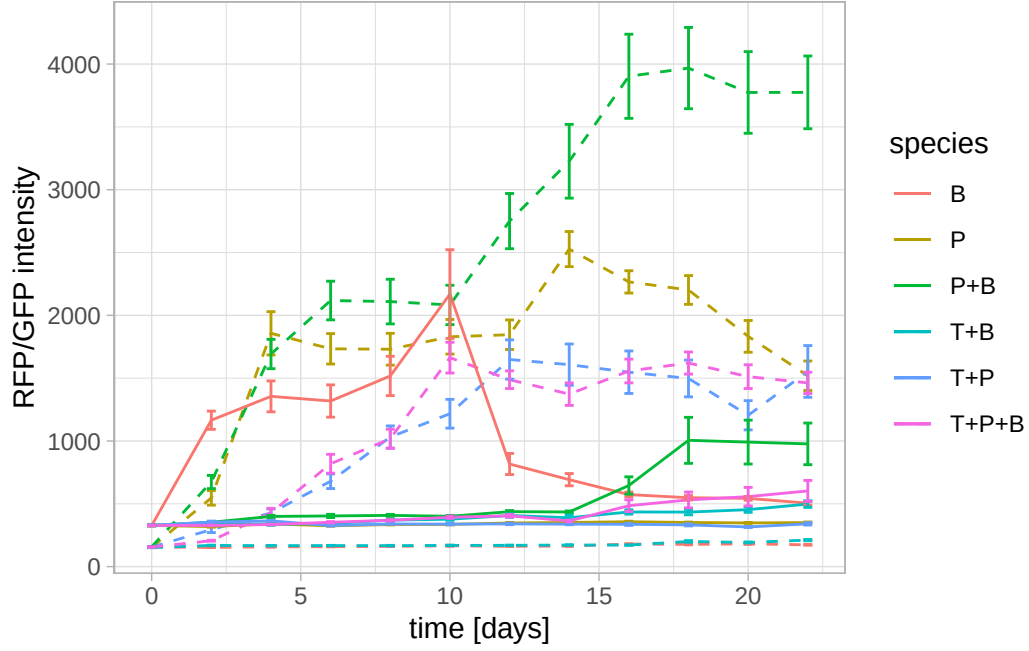


Figure 2: Time series growth data for all different mono/co-culturing combinations of *B. subtilis* (RFP), *P. putida* (GFP) and *T. viride* (no growth data) in a micro-scale maze with fractal complexity of f3 (Hilbert curve) on microfluidic chips. Solid lines indicate RFP signal, dashed lines indicate GFP signal.

4.2 Parametrization of Lotka—Volterra interaction coefficients suggests shift in intra- and inter-specific competition with increased spatial complexity

The parameters obtained from the per-capita growth regression and subsequent optimization (Methods, 3.2) were used to predict growth curves for all conditions and fractal levels (Figure 3). *P. putida* monocultures reach a clear carrying capacity, with slightly reduced growth at the highest fractal level (f5). When co-cultured with *T. viride*, *P. putida* shows a lagged onset of growth. In the presence of *B. subtilis*, *P. putida* has a prolonged logistic growth phase, and it is unclear whether the carrying capacity is reached within the experimental period. The predicted growth pattern of the three-species culture is similar to that of the *P. putida* + *T. viride* culture. The LV model fits the *B. subtilis* monoculture poorly ($R^2 = 0.133$ – 0.169 , Table 2), whereas all other conditions are fitted well ($R^2 > 0.83$).

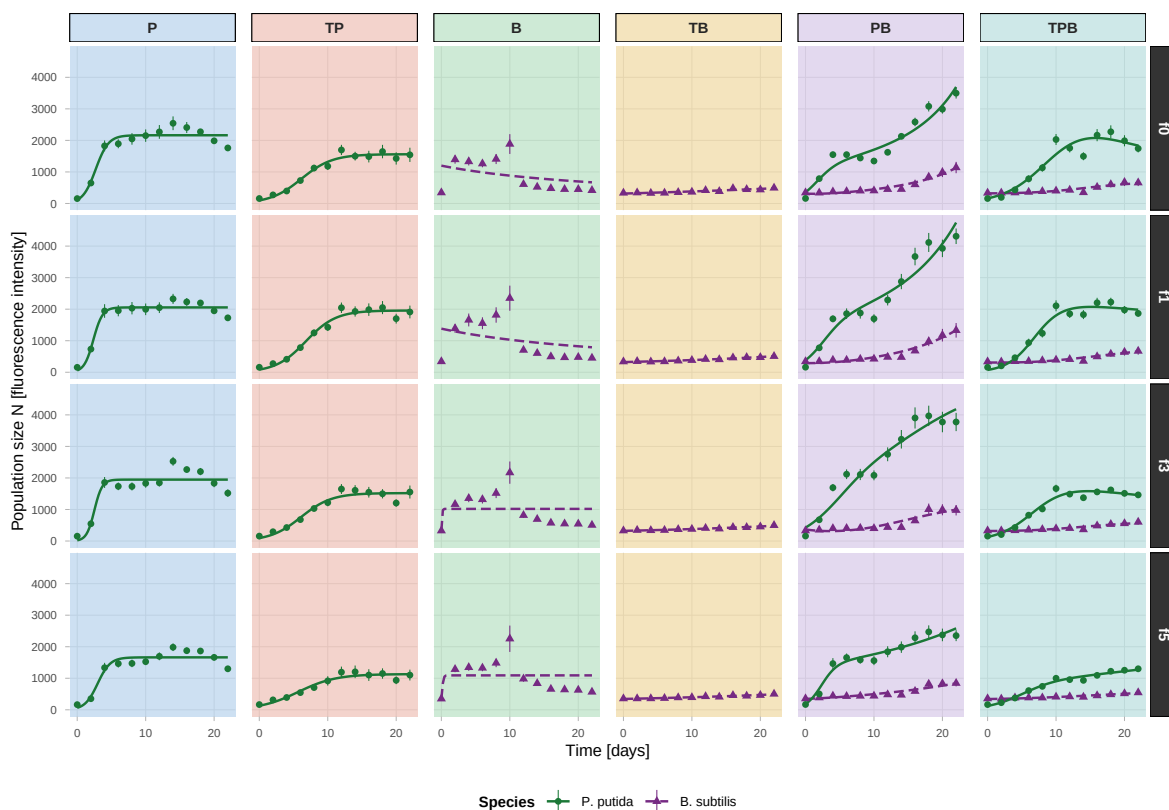


Figure 3: Observed (points, $\pm$ SE) and Lotka—Volterra predicted (lines) growth curves across fractal levels (rows) and conditions (columns).

The interaction coefficients are summarized in a heatmap (Figure 4), where hue indicates the coefficient and color saturation the R^2 of the fit for the species whose growth equation contains the coefficient. Full R^2 values are given in the supplementary material (Table 2). *B.*

subtilis shows strong self-inhibition (a_{22}) at higher fractal complexity in monoculture, which is mitigated when it is co-cultured with *P. putida*. Self-inhibition of *B. subtilis* is also substantially lower in co-culture with *T. viride*. In contrast, self-inhibition of *P. putida* (a_{11}) changes little between monoculture and any of the co-cultures. The coefficients suggest a promoting effect of *B. subtilis* on *P. putida* in the two-species culture, which increases with spatial complexity. This effect shifts when *T. viride* is added: in the three-species system, a_{12} tends to be negative (inhibitory) at low complexity and mediocre complexity, but changes to a positive effect on at the highest spatial complexity (f5).

Overall all interaction coefficients are values quite close to 0.

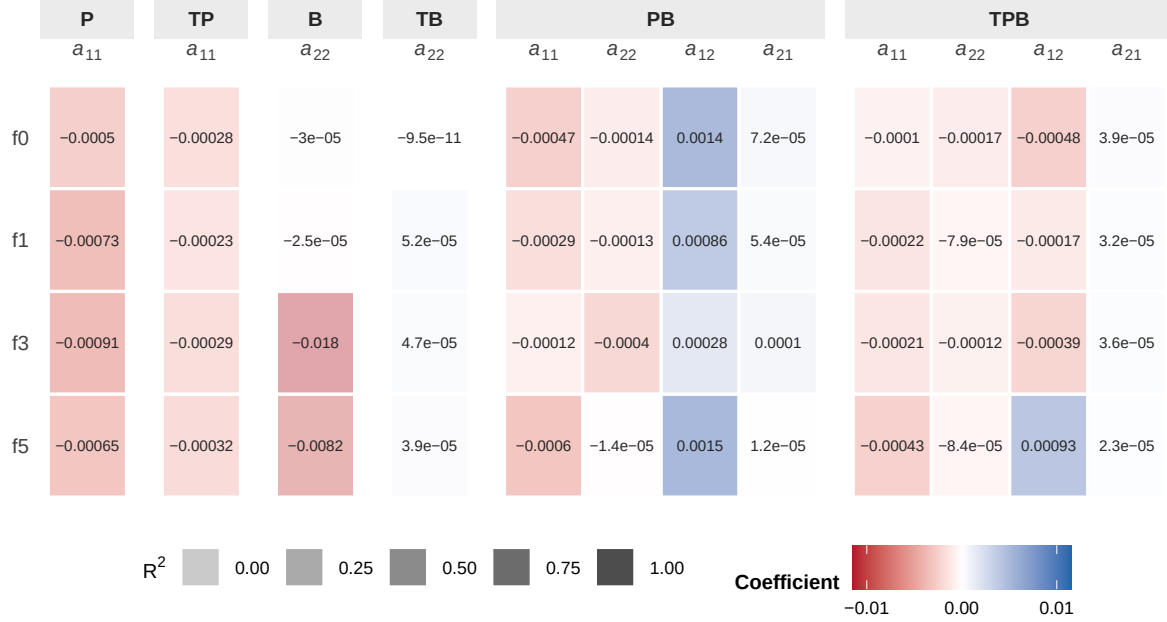


Figure 4: Interaction coefficients per fractal level (rows) and condition (columns). Tile colour shows the estimated coefficient (blue: positive, red: negative); colour saturation shows R^2 of the fit of the species whose growth equation contains the coefficient (faint = poor fit). Numbers are the coefficient values.

5 Discussion

The aim of this project was to test whether an LV model can describe growth data of a three-species community when only two species are measured and the third acts as a perturbation of their interaction. The model fitted most conditions well, but several results limit how far the coefficients can be interpreted mechanistically. The main points are discussed below, together with assumptions and weaknesses of the approach.

5.1 Poor fit of the *B. subtilis* monoculture

The optimizer could not fit the *B. subtilis* monoculture ($R^2 = 0.13$ — 0.17 , Table 2), which is the most notable failure of the model. A likely reason is that the growth curve deviates from the logistic shape assumed by the model, since the species abundance decreased after a peak and a death phase, or more general, a transient peak followed by decline, is not supported by the model which only fits single logistic curves. The poor reflection could also be aggravated by the comparable small sample size ($n = 7$) for the *B. subtilis* monocultures.

This also affects the reliability of the strong inter-specific competition (a_{22}) found in the dataset (-0.018 at f3, versus way closer to 0 elsewhere) which might therefore rather be an artefact than reproducible biological effect.

5.2 Interpretation of interaction coefficients

The promoting effect of *B. subtilis* on *P. putida* seen in the growth curves is well reflected by the interaction coefficients (a_{12} in the PB condition, Figure 4). In contrast, the lowered self-inhibition of *B. subtilis* in co-culture with *T. viride* does not reflect the strong inhibitory effect of the fungus on *B. subtilis*. Modeling the two-species co-culture from the single-culture readout alone is therefore poor in this case.

In the *T. viride* + *B. subtilis* condition, *B. subtilis* hardly grows, so its per-capita growth rate is flat and close to zero across the observed abundance range. A near-zero a_{22} here then reflects the absence of growth, not release from self-limitation. The coefficient can only quantify self-limitation when growth occurs on a properly measurable range.

The effect of *P. putida* on *B. subtilis* (a_{21}) is close to zero in all co-cultures, even though the observed growth data shows strong suppression of *B. subtilis* by *P. putida*. This is poorly reflected by the chosen modeling approach.

In the three-species condition, the fitted a_{12} and a_{21} are effective coefficients that combine direct effects with effects mediated by *T. viride*. They are not comparable to the coefficients from the two-species culture in a strict sense, and they cannot distinguish pairwise competition from higher-order interactions. This is illustrative for the general limitation of the approach: the effect of the unmeasured species cannot be simply assigned to an explicit interaction term and is folded into the parameters of the measured species.

Reflecting on the modeling framework on a broader scope, the optimizer works in log-space and fixes the sign of every parameter to that of the initial linear estimate, so a wrongly signed starting value cannot be corrected. Poorly constrained, near-zero coefficients may therefore switch sign between fits without real biological meaning.

As an outlook for further work the estimated parameters could be used to simulate the growth trajectories of the co-cultures over longer periods, to test whether stable coexistence or cycling patterns would occur.

6 Statement of authorship, contributions and use of LLMs

This report is submitted as the practical project for the class “Practical Course in Modelling and Systems Biology” (2026S 300332-1). I originally wanted to conduct the modeling analyses with my own masters thesis data from co-culturing experiments with *P. putida*, *B. subtilis* and a complex soil bacterial community isolated from soil samples as a natural inoculum. However, since I was unable to have the necessary experimental data ready in time for the due date of this report I herein use the data of my colleague Maryam Al-Jalahwee, BSc who conducted similar experiments with *Trichoderma viride* as the third interaction entity. All data arranging, sorting and subsequent analysis has been conducted by the author.

I used CLAUDE *Sonnet 5* and OpenAI GPT-5.6 *Luna* for generation and improvements of R code. In particular I used it for the building of functions to automate my data processing pipeline for all the different conditions and fractal levels I have in my data, for bug fixing, efficiency improvements on the code I provided and for configuring layout, design and arrangement of plots and tables. I also used it to help me better understand the mechanism of some functions I used.

7 References

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8 Supplementary Material

The complete code that contains all data handling and also compiles this report reproducibly can be found as a Quarto document [GitHub](#).

Table 1: Estimated Lotka-Volterra interaction coefficients for cocultures of *P. putida* (1) and *B. subtilis* (2) on microfluidic chips with mazes of increasing fractal complexity (f0-f5). a_{11} and a_{22} are the intra-specific effects of *P. putida* and *B. subtilis*, respectively; a_{12} and a_{21} are the inter-specific effects of *B. subtilis* on *P. putida* and vice versa. Negative coefficients indicate competition/inhibition, positive coefficients promotion/enhancement.

Fractal	<i>P. putida</i>		<i>T. viride</i> + <i>P. putida</i>		<i>B. subtilis</i>		<i>T. viride</i> + <i>B. subtilis</i>		<i>P. putida</i> + <i>B. subtilis</i>		<i>T. viride</i> + <i>P. putida</i> + <i>B. subtilis</i>	
	a_{11}	a_{22}	a_{11}	a_{22}	a_{11}	a_{22}	a_{11}	a_{22}	a_{11}	a_{22}	a_{11}	a_{21}
f0	-5e-04	0.0000	-3e-04	0.0000	-5e-04	-1e-04	0.0014	1e-04	-1e-04	-2e-04	-5e-04	0
f1	-7e-04	0.0000	-2e-04	0.0000	-3e-04	-1e-04	0.0009	1e-04	-2e-04	-1e-04	-2e-04	0
f3	-9e-04	-0.0181	-3e-04	0.0000	-1e-04	-4e-04	0.0003	1e-04	-2e-04	-1e-04	-4e-04	0
f5	-7e-04	-0.0082	-3e-04	0.0000	-6e-04	0e+00	0.0015	0e+00	-4e-04	-1e-04	9e-04	0

Table 2: Goodness of fit of the fitted growth curves against the observed means, per species. $R^2 = 1 - SS_{\text{res}}/SS_{\text{tot}}$ (can be negative for fits worse than the mean); RMSE in the units of N (fluorescence intensity). Model predictions were interpolated at the observed time points.

Condition	Fractal	<i>P. putida</i>		<i>B. subtilis</i>	
		R^2	RMSE	R^2	RMSE
<i>P. putida</i>	f0	0.913	201.964	–	–
	f1	0.942	150.121	–	–
	f3	0.836	263.606	–	–
	f5	0.885	186.036	–	–
<i>T. viride</i> + <i>P. putida</i>	f0	0.966	99.799	–	–
	f1	0.969	124.024	–	–
	f3	0.942	126.850	–	–
	f5	0.936	90.757	–	–
<i>B. subtilis</i>	f0	–	–	0.169	466.131
	f1	–	–	0.137	608.991
	f3	–	–	0.133	487.614
	f5	–	–	0.161	467.490
<i>T. viride</i> + <i>B. subtilis</i>	f0	–	–	0.845	22.368
	f1	–	–	0.951	13.095
	f3	–	–	0.956	10.777
	f5	–	–	0.943	11.091
<i>P. putida</i> + <i>B. subtilis</i>	f0	0.935	240.730	0.956	55.194
	f1	0.929	343.371	0.949	75.043
	f3	0.930	323.308	0.861	95.776
	f5	0.954	148.544	0.880	60.509
<i>T. viride</i> + <i>P. putida</i> + <i>B. subtilis</i>	f0	0.909	225.997	0.865	45.482
	f1	0.960	151.623	0.896	38.554
	f3	0.959	110.466	0.904	28.892
	f5	0.979	54.865	0.933	15.762

(continued)

Table 3: Parameter estimates with uncertainty for every fit. r_i : intrinsic growth rate of species i ; a_{ij} : interaction coefficients (indices as in Table 1). SD is half the width of the ± 1 SD interval returned by `gauser`; the 95% interval is the normal approximation (estimate ± 1.96 SD). ‘Excludes 0’ indicates whether this interval excludes zero.

Condition	Fractal	Parameter	Estimate	SD	95% interval	Excludes 0
<i>P. putida</i>	f0	P_0	114.2127	3.9e+04	-7.6e+04, 7.6e+04	no
		r_1	1.0713	5.8	-10, 12	no
		a_{11}	-0.0005	0.0029	-0.0061, 0.0051	no
	f1	P_0	58.4481	7.1e+05	-1.4e+06, 1.4e+06	no
		r_1	1.4975	16	-30, 33	no
		a_{11}	-0.0007	0.0082	-0.017, 0.015	no
	f3	P_0	22.7074	7.4e+06	-1.5e+07, 1.5e+07	no
		r_1	1.7716	27	-50, 54	no
		a_{11}	-0.0009	0.015	-0.029, 0.028	no
	f5	P_0	62.4204	1.8e+05	-3.5e+05, 3.5e+05	no
		r_1	1.0877	8.9	-16, 18	no
		a_{11}	-0.0007	0.0058	-0.012, 0.011	no
<i>T. viride</i> + <i>P. putida</i>	f0	P_0	97.2307	69	-38, 2.3e+02	no
		r_1	0.4414	0.15	0.14, 0.74	yes
		a_{11}	-0.0003	0.00012	-0.00053, -4e-05	yes
	f1	P_0	84.7892	2.2e+02	-3.4e+02, 5.1e+02	no
		r_1	0.4556	0.31	-0.16, 1.1	no
		a_{11}	-0.0002	0.0002	-0.00062, 0.00015	no
	f3	P_0	93.9718	9.5e+02	-1.8e+03, 2e+03	no
		r_1	0.4456	0.57	-0.68, 1.6	no
		a_{11}	-0.0003	0.00043	-0.0011, 0.00055	no
	f5	P_0	129.0483	87	-41, 3e+02	no
		r_1	0.3608	0.23	-0.097, 0.82	no
		a_{11}	-0.0003	0.00028	-0.00087, 0.00023	no
<i>B. subtilis</i>	f0	B_0	1199.2839	3.7e+02	4.8e+02, 1.9e+03	yes
		r_2	0.0000	Inf	-Inf, Inf	no
		a_{22}	-0.0000	3e-05	-8.9e-05, 2.9e-05	no
	f1	B_0	1382.1754	4.2e+02	5.7e+02, 2.2e+03	yes
		r_2	0.0000	Inf	-Inf, Inf	no
		a_{22}	-0.0000	2.5e-05	-7.3e-05, 2.4e-05	no
	f3	B_0	327.2705	1.3e+03	-2.2e+03, 2.8e+03	no
		r_2	18.3931	Inf	-Inf, Inf	no
		a_{22}	-0.0181	Inf	-Inf, Inf	no
	f5	B_0	350.1735	1.4e+03	-2.4e+03, 3.1e+03	no
		r_2	8.9568	4.1e+78	-8e+78, 8e+78	no
		a_{22}	-0.0082	3.8e+75	-7.4e+75, 7.4e+75	no

(continued)

Condition	Fractal	Parameter	Estimate	SD	95% interval	Excludes 0
<i>T. viride</i> + <i>B. subtilis</i>	f0	B_0	314.1291	1.4e+02	37, 5.9e+02	yes
		r_2	0.0194	0.044	-0.066, 0.11	no
		a_{22}	-0.0000	Inf	-Inf, Inf	no
	f1	B_0	322.0550	1.3e+02	70, 5.7e+02	yes
		r_2	0.0000	Inf	-Inf, Inf	no
		a_{22}	0.0001	0.00011	-0.00016, 0.00026	no
	f3	B_0	322.1593	1.3e+02	68, 5.8e+02	yes
		r_2	0.0000	Inf	-Inf, Inf	no
		a_{22}	0.0000	0.00012	-0.00018, 0.00028	no
	f5	B_0	344.2782	1.4e+02	74, 6.1e+02	yes
		r_2	0.0000	Inf	-Inf, Inf	no
		a_{22}	0.0000	0.00013	-0.00022, 0.00029	no
<i>P. putida</i> + <i>B. subtilis</i>	f0	B_0	310.5821	2.2e+02	-1.2e+02, 7.4e+02	no
		P_0	353.4461	1.5e+05	-3e+05, 3e+05	no
		r_1	0.2093	2.5e+02	-4.9e+02, 4.9e+02	no
		r_2	0.0020	1.8e+02	-3.5e+02, 3.5e+02	no
		a_{11}	-0.0005	3.8	-7.5, 7.5	no
		a_{22}	-0.0001	0.00078	-0.0017, 0.0014	no
		a_{12}	0.0014	9.1	-18, 18	no
		a_{21}	0.0001	0.00012	-0.00017, 0.00032	no
	f1	B_0	289.3245	1.3e+02	28, 5.5e+02	yes
		P_0	381.6527	8.4e+02	-1.3e+03, 2e+03	no
		r_1	0.2973	1.6	-2.8, 3.4	no
		r_2	0.0165	0.088	-0.16, 0.19	no
		a_{11}	-0.0003	0.0048	-0.0097, 0.0091	no
		a_{22}	-0.0001	0.00055	-0.0012, 0.00095	no
		a_{12}	0.0009	0.014	-0.027, 0.028	no
		a_{21}	0.0001	7.3e-05	-9e-05, 0.0002	no
	f3	B_0	367.1730	3.5e+02	-3.2e+02, 1.1e+03	no
		P_0	425.9024	6e+02	-7.5e+02, 1.6e+03	no
		r_1	0.2497	0.28	-0.29, 0.79	no
		r_2	0.0189	2	-3.9, 3.9	no
		a_{11}	-0.0001	0.00057	-0.0012, 0.001	no
		a_{22}	-0.0004	0.00025	-0.00089, 8.2e-05	no
		a_{12}	0.0003	0.0074	-0.014, 0.015	no
		a_{21}	0.0001	5.9e-05	-1.6e-05, 0.00022	no
	f5	B_0	346.1842	3e+02	-2.5e+02, 9.4e+02	no
		P_0	185.5721	1.1e+04	-2.1e+04, 2.2e+04	no
		r_1	0.3586	1.3e+02	-2.6e+02, 2.6e+02	no
		r_2	0.0271	0.1	-0.17, 0.22	no
		a_{11}	-0.0006	0.0078	-0.016, 0.015	no
		a_{22}	-0.0000	0.015	-0.03, 0.03	no
		a_{12}	0.0015	0.3	-0.59, 0.59	no
		a_{21}	0.0000	0.0002	-0.00039, 0.00041	no

(continued)

Condition	Fractal	Parameter	Estimate	SD	95% interval	Excludes 0
<i>T. viride</i> + <i>P. putida</i> + <i>B. subtilis</i>	f0	B_0	320.5456	2.5e+02	-1.7e+02, 8.1e+02	no
		P_0	160.2818	1.1e+04	-2.1e+04, 2.2e+04	no
		r_1	0.4678	0.52	-0.54, 1.5	no
		r_2	0.0520	0.32	-0.57, 0.67	no
		a_{11}	-0.0001	0.48	-0.95, 0.95	no
		a_{22}	-0.0002	0.0051	-0.01, 0.0097	no
		a_{12}	-0.0005	0.09	-0.18, 0.18	no
	f1	a_{21}	0.0000	0.001	-0.0019, 0.002	no
		B_0	308.3955	2.4e+02	-1.6e+02, 7.8e+02	no
		P_0	72.9189	1.5e+02	-2.2e+02, 3.7e+02	no
		r_1	0.5479	0.3	-0.044, 1.1	no
		r_2	0.0213	11	-22, 22	no
		a_{11}	-0.0002	0.0005	-0.0012, 0.00076	no
		a_{22}	-0.0001	0.043	-0.085, 0.084	no
	f3	a_{12}	-0.0002	0.16	-0.32, 0.32	no
		a_{21}	0.0000	0.00064	-0.0012, 0.0013	no
		B_0	318.1260	2.8e+02	-2.3e+02, 8.6e+02	no
		P_0	128.4779	1e+03	-1.9e+03, 2.2e+03	no
		r_1	0.5132	0.73	-0.92, 1.9	no
		r_2	0.0355	9	-18, 18	no
		a_{11}	-0.0002	0.00092	-0.002, 0.0016	no
	f5	a_{22}	-0.0001	0.053	-0.1, 0.1	no
		a_{12}	-0.0004	0.012	-0.023, 0.022	no
		a_{21}	0.0000	0.0074	-0.014, 0.015	no
		B_0	341.1477	1.7e+02	15, 6.7e+02	yes
		P_0	122.7393	1e+03	-1.9e+03, 2.1e+03	no
		r_1	0.0742	50	-98, 98	no
		r_2	0.0364	12	-23, 23	no
		a_{11}	-0.0004	0.002	-0.0043, 0.0035	no
		a_{22}	-0.0001	0.00045	-0.00097, 0.00081	no
		a_{12}	0.0009	0.0087	-0.016, 0.018	no
		a_{21}	0.0000	4.4	-8.6, 8.6	no